Tasks 2. Mixture

Model the following random variables with N=10000, which from a mixture of some distributions (using both methods), and compare the theoretical moments (math.expectation and variance) with empirical ones. And compare their plots for distribution functions and density functions. And calculate the difficulty of the algorithm for both cases.

1. Simulate a mixture of the two following distributions. One is normal with parameters mean = 1, variance = 9; the other with the following density f(x) with $\alpha = 3$ and $\beta = 2$. The mixture weight p is 0.4.

$$f(x) = \frac{\alpha}{2} e^{-\alpha|x-\beta|}$$

2. The random variable X has the following probability density function:

$$f(x) = \begin{cases} a\,(1-x^2), & -1 \le x \le 1, \\ \dfrac{1}{2}\,e^{-(x-2)}, & x \ge 2, \\ 0, & \text{otherwise.} \end{cases}$$

3. Consider a function $f(x)$ consisting of three components, one of which is constant and equal to $\frac{1}{4}$ on the interval $[1, 2)$.

$$f(x) = \begin{cases} \dfrac{1}{8}(x+1), & -1 \le x < 1, \\ \dfrac{1}{4}, & 1 \le x < 2, \\ \dfrac{1}{2} e^{-(x-2)}, & x \ge 2, \\ 0, & \text{otherwise.} \end{cases}$$